Ensieh Habibi

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Professional Summary

Research Scientist and molecular biologist with integrated wet- and dry-lab expertise in next-generation sequencing and multi-omics. Experienced in genomic and epigenomic data analysis, including quality control, genotyping, DNA methylation profiling, co-methylation network analysis, chromatin state annotation, workflow automation, and large-scale integration of single-cell and bulk datasets. Brings a multidisciplinary perspective at the intersection of genomics, epigenetics, and environmental influences on development, with a strong focus on translating biological insights into clinical and diagnostic applications.

Professional Experience

Postdoctoral Fellow - UC Davis | 2022- Present
LaSalle Lab, Department of Microbiology & Immunology - 2023-Present

- Project1: Single-cell methylation profiling of rhesus macaque oocytes to identify stress-associated epigenetic biomarkers and traced inheritance patterns into embryos to study epigenetic disease risk
- Project2: hydroxymethyl analysis of the imprinted *Snord116* region in mouse cortex to study light- and sex-dependent regulation of circadian and metabolic phenotypes.
- Project3: epigenetic and transcriptomic profiling of the autism-associated *NHIP* region across rhesus macaque samples from distinct physiological conditions: stress (cumulus), obesity (brain and cfDNA), smoked exposure (nasal).

Genomic Variation Lab - 2022 - 2023

- Project: Population-scale genomic analysis of zooplankton to assess genetic diversity and connectivity across freshwater ecosystems.

Research Intern - Stanford University | 2021
Snyder Lab, Department of Genetics

- Project: RNA-seq-based eQTL analysis in colon cancer to link genetic variants with expression regulation.

Graduate Researcher - UC Davis | 2017–2022
Miller Lab & Genomic Variation Lab

- Project: Genomic and epigenomic analyses of endangered fish to design population specific SNP assay, identify genetic variation, markers of selection and adaptation, and transgenerational inheritance of methylation patterns over 4 generations.

Research Assistant - TU Dresden, Germany | 2015–2016
Reinhardt Lab, Department of Applied Zoology

- Project: Mito-nuclear genome interaction effects on fertility and oxidative stress in *Drosophila melanogaster.*

Technical Skills

Dry Lab Techniques:

- **Next-Generation Sequencing (NGS):**
  scWGBS, WGBS, WGS, RNA-seq, ChIP-seq, RAD-seq
- **Programming & Workflow Automation:**
  Python, R, Bash; Snakemake, Conda, Git, Jupyter Notebook
- **Data Analysis & Computational Biology:**
  Differential methylation (DMR/DMG), clustering, co-expression and co-methylation network analysis, chromatin state

annotation, TF regulatory network analysis, SNP assay development, population genetics (Fst, PCA, Admixture), and machine learning (Random Forest, SVM on DMRs using DMRichR)

- **High-Performance Computing (HPC):**
  Extensive experience running large-scale sequencing data on SLURM-based Unix/Linux clusters; parallelized and automated workflows for multi-sample datasets
- **Epigenomics & Single-Cell Pipelines:**
  AllCools, DMRichR, CoMethyl, CpG_Me, Rocketchip, hdWGCNA, Scanpy, Seurat
- **Genomics Tools:**
  BWA, Samtools, Bedtools, FastQC, Picard, Bismark, Bowtie2, ANGSD, HOMER, ChromHMM
- **Databases & Resources:**
  UCSC Genome Browser, NCBI, ChIP-Atlas, ENCODE, TRUST

Wet Lab Techniques:

- **NGS Library Preparation & QC:** Experienced in preparing and validating libraries for scWGBS, WGBS, WGS, and RAD-seq using Bioanalyzer, TapeStation, Qubit
- **Molecular Biology:** Skilled in DNA/RNA extraction, PCR/qPCR, gel electrophoresis (agarose and acrylamide), and AFLP electrophoresis for fragment analysis and sequencing QC

Selected Publications

- **Habibi, E.**, Zarrabi, J., Hernandez, C., Torres, C., Su, V., von Behran, Z., Tsai, M., Zhang, Y., Luo, C., Tracredi, D., Walker, C., VandeVoort, C. & LaSalle, J. *Maternal stress prior to conception alters DNA methylation in primate oocytes. In review - Nature Communication 2025*
- Williams AL, **Habibi E,** Angel E, Schmidt RJ, LaSalle JM. *Prenatal cell-free DNA methylome detects association with autism and maternal obesity. In review - Communication Biology* 2025
- **Habibi E,** Miller MR, Finger AJ, Andrea Schreier. *Single generation epigenetic adaptation to captivity and reinforcement in subsequent generations in a delta smelt (Hypomesus transpacificus) conservation hatchery. Molecular Ecology, 2024*
- **Habibi E,** Miller MR, Gille G, Sanders L, Rodzen J, Stephens M, Finger AJ. *Genetic monitoring of Upper McCloud River Redband Trout. Conservation Genetics, 2022*
- Campbell MA, **Habibi E,** Auringer G, Stephens M, Rodzen J, Finger AJ. *Polyphyly of the Sacramento Redband Trout Oncorhynchus mykiss stonei and the description of the McCloud River Trout Oncorhynchus mykiss winnememi, new subspecies. Zootaxa, 2021*

Education

- **Ph.D. in Animal Biology**, University of California, Davis | 2017 – 2022
  Dissertation: *Genomics and Epigenomics Tools for Conservation Genetics*
- **M.S.,** Tarbiat Modares University, Iran | 2007 – 2010

Awards & Fellowships

- Ursula Abbott Travel Award, UC Davis – $1,500 (2022)
- Henry A. Jastro Graduate Research Award, UC Davis – $3,000 (2020)
- Ursula Abbott Travel Award, UC Davis – $2,000 (2019)
- Marin Rod & Gun Club Scholarship – $2,000 (2019)
- Animal Biology Graduate Group Fellowship – $1,800 (2018)
- Top Student in Class of 2007 (out of 50 students) – 2007

Advisory & Intellectual Property

- Genetics & Genomics Technical Advisor, TRIP Research (Nonprofit)
- Invention Disclosure, University of California
  Diagnostic for Detecting Preconception Stress from Oocytes and Cumulus Cells, Role: Inventor